

Q: What is Oracle 12c database all about?

Oracle Database 12c has been designed to meet the needs of the cloud-based environment, providing a new multi-tenant architecture on top of a fast, scalable, reliable, and secure database platform. It has 500 additional features, and can greatly benefit customers deploying private database clouds as well as Software-as-a-Service (SaaS) vendors looking for the power of Oracle Database in a secure multi-tenant model. Oracle Multi-tenant works with all Oracle Database features, including real application clusters, partitioning, Data Guard, compression, automatic storage management, real application testing, transparent data encryption, Database Vault, and more.

Oracle Database 12c customers can accelerate performance of Oracle Multi-tenant pluggable databases by using the fastest NAS database performance available from the Sun ZFS Storage Appliance. The Oracle Intelligent Storage Protocol (OISP) introduced in the database will enable it to communicate directly with the Sun ZFS Storage Appliance, and deliver a 65 per cent reduction in administration and provisioning by dynamically and automatically tuning database performance. OISP will allow select future Oracle storage products to highly optimise configurations, accelerate provisioning, and auto-tune storage systems for increased performance. Also, to help customers efficiently manage more data, lower storage costs and improve database performance, Oracle Database 12c brings along the new Automatic Data Optimisation feature. Available with Oracle Advanced Compression, this feature automatically balances and optimises performance by reducing the storage footprint to the maximum extent using the massive data compression uniquely available on Oracle storage systems—Hybrid Columnar Compression (HCC).

Q: What do you mean by a database designed for the cloud?

Earlier, cloud service providers were trying to provide one application hosted on a single server to multiple customers. It is pretty similar to the way electricity is distributed. The challenge with data is that at an application level, you might have it being given to different customers. But when the data goes into the server, it starts co-mingling. So the privacy of data is compromised. Although there are ways of stopping this, they are not fool proof. The only fool proof way is if the data of different customers is kept separate, physically. Because of this challenge, a lot of large companies were not moving to the cloud despite the fact that it is cheaper to do so. That is where Oracle 12c makes the difference. What we have tried doing here is to ensure that data of different companies can be kept separate because we will only have a pluggable database for everyone. So the security and privacy of a database is well taken care of. I can confidently say that the security threat on the cloud is now passé, because there can be no intermingling of data anymore.

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Q: There are many software companies that want to become SaaS vendors. How will the multi-tenancy architecture help them?

The companies that want to become SaaS vendors want multi-tenancy and they know how to go for it. They would not get customers if they provided the software at an application level because they will not be able to manage the database separately. Today, with multi-tenancy at the database, it is easier for them to get more customers. So with 12c, they can get even more customers. Let's look at the example of rural and co-operative banks going in for computerisation. Each bank will not have enough money to set up its own separate data centre. Wouldn't it be nice if there was one SaaS vendor who could host the data for all these banks separately? With 12c, the challenges these banks have been facing related to data security and privacy have been taken care of.

Q: Who do you think will be the users of this cloud database?

Everyone who has multiple databases to manage can use a cloud database. So companies that have been managing different databases in different servers can now manage these together in one server with 12c, saving a lot on the cost of maintaining and managing servers. The beauty of it is that, tomorrow, if you want to modify a pluggable database into a standard database, you can do it.

Q: For large companies that already have a private cloud, how does 12c help?

A lot of large companies nowadays prefer to have a private cloud, so 12c makes it far easier for them to administer things and drive better performance while reducing their costs.